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Characterizing preschooler's outdoor physical activity: The comparability of schoolyard location- and activity type-based approaches

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ABSTRACT

Understanding the context of children's schoolyard activity can inform approaches by which to increase physical activity participation. Two major approaches exist for assessing this context- based on location or activity type. Location-based methods (e.g., Global Positioning Systems) may miss variations in activity type if children participate in multiple types of activities in each location (e.g., children engage in something other than playing with sand in a sandbox). Activity-type based methods for contextualizing activity (e.g., Observational System for Recording Physical Activity in Children, OSRAC) capture activity type, but not location, and children might participate in the same type(s) of activities in multiple schoolyard locations. The purpose of this study was to compare location- vs activity type-based approaches to characterizing children's schoolyard behavior. Preschoolers' ($N = 50$) schoolyard location and activity type (based on OSRAC) were video-coded and matched with physical activity monitor data (counts/s) for 1 outdoor period (~43 min). The results showed that multiple activities occurred within each location (median = 5), but the predominant activity typically matched the expected activity type (e.g., 66.9% fixed equipment play in the fixed equipment location). Children spent the majority of their time in open space (type or location; 37.6%–48.9%), but there were differences in the setting that promoted the most physical activity (teacher arranged type, but fixed equipment location). Activity types elicited varying physical activity levels depending on location (e.g., open space activities in open spaces vs on paths: 40.9 vs 60.7 counts/s). Activity type- and location-based approaches provide some similar, but potentially additive information about schoolyard behavior, which is an important methodological consideration for future studies.

Introduction

There is continued interest in how the schoolyard physical environment impacts young children's behavior, as outdoor time is a crucial opportunity for children to engage in physical activity in addition to motor, cognitive, and social development through exploration and free-play (Cosco, Moore & Islam, 2010; Hustyi, Normand, Larson & Morley, 2012; Lerstrup & van den Bosh, 2017; Nicaise, Kahan & Sallis, 2011, 2012; Sevimli-Celik, 2018; Smith et al., 2014; Storli & Hagan, 2010; Wishart et al., 2019; Wyver & Little, 2018). Information about the schoolyard's impact on physical activity can be used to inform activity-promoting schoolyard redesign interventions (Andersen et al., 2015b; Nicaise et al., 2012; Raney, Hendry & Yee, 2019), which may be an integral part of a population-wide strategy to combat obesity and inactivity (Gebel, Bauman & Petticrew, 2007).

Prior studies have characterized preschoolers' schoolyard behavior by assessing either the type of activity or play in which a child is engaged (Brown et al., 2009; Howie, Brown, Dowda, McIver & Pate, 2013; Nicaise et al., 2011, 2012; Soini et al., 2014; Wishart et al., 2019) or the child's location on the schoolyard (Clevenger et al., 2020a; Cosco et al., 2010; Holmes, 2012; Holmes & Procaccino, 2009; Nicaise et al., 2011, 2012; Wishart et al., 2019). While the theory of affordances (Gibson, 1977) supports that activity type and location are related- features of the schoolyard's physical environment (location) impact the types of activities in which children engage- the 2 terms are inherently not interchangeable. Understanding the similarities and differences between activity type- and location-based approaches to characterizing school-

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yard physical activity is essential for comparing the results of prior studies and designing future trials.

Theoretical framework

Gibson's theory of affordances provides a theoretical basis supporting the environment's influence on behavior, positing that we perceive our environment as opportunities for action and this subsequently drives our actions (e.g., if children perceive that climbing is afforded, they will climb) (Gibson, 1977). In the schoolyard context, different areas of the schoolyard inherently afford different types of activities (e.g., large, open areas afford running games), which impacts physical activity participation (Fjørtoft, Kristoffersen & Sageie, 2009; Lerstrup & van den Bosch, 2017). Heft (1988) provided a means by which affordances (e.g., running, skipping, throwing) of children's play areas (e.g., flat surfaces, smooth slopes) could be categorized, a taxonomy that was later expanded by Kyttä (2002) to include affordances for sociality, like "affords rule games" and updated by Lerstrup and van den Bosch (2017). However, Kyttä (2004) notes an important distinction between potential and actualized affordances. An area of the schoolyard may afford any number of activity types (potential affordances), but children may only utilize a small number of these afforded activities (actualized affordances) (Kyttä, 2004) due to personal characteristics (e.g., age or experience in that environment) (Cosco, 2006; Kyttä, 2004) or features of the environment (Storli & Hagen, 2010).

Physical activity in child care and on the schoolyard

The Institute of Medicine recommends that children participate in at least 15 min/h of total physical activity, defined as any bodily movement that increases energy expenditure above a basal level (Piercy et al., 2018), while in childcare, including light, moderate, and vigorous intensities (McGuire, 2012). Approximately half of children meet this guideline (Pate et al., 2015), but there is a high degree of variability between child care programs (Gubbels et al., 2011; Pate, McIver, Dowda, Brown & Addy, 2008; Tandon, Saelens, Zhou, Kerr & Christakis, 2013; Vanderloo et al., 2014), which may be partially attributed to environmental or contextual differences (Henderson, Grode, O'Connell & Schwartz, 2015).

In the childcare setting, outdoor time is of particular interest because it provides children the opportunity for unstructured (i.e., free-play) physical activity during the otherwise predominantly structured or sedentary day. Unfortunately, children are still mostly sedentary during outdoor time (53% of time according to a recent review; Truelove et al., 2018), despite being more active outdoors compared to indoors (Brown et al., 2009; Raustorp et al., 2012; Tandon et al., 2013; Vanderloo, Tucker, Johnson & Holmes, 2013). Several features of schoolyard design have been associated with physical activity participation, although results are mixed (Broekhuizen, Scholten & De Vries, 2014). For example, moderate-to-vigorous physical activity has been related to portable (Berg, 2015; Bower et al., 2008; Dowda et al., 2009; Gunter, Rice, Ward & Trost, 2012) or fixed (Bower et al., 2008; Gunter et al., 2012; Sugiyama, Okely, Masters & Moore, 2012) equipment, but other studies report no effect or a negative association (Cardon, Van Cauwenberghe, Labarque, Haerens & De Bourdeaudhuij, 2008; Copeland, Khoury & Kalkwarf, 2016; Dowda et al., 2009). Findings may be inconsistent because researchers have correlated the overall presence or absence of particular features of the outdoor environment to overall physical activity levels, an approach that cannot inform us as to children's physical activity levels while in a particular area or while participating in a particular type of activity.

As evidenced by observational studies examining the physical activity of preschool-aged children in different schoolyard locations, it is clear that physical activity participation varies by location on the schoolyard (Cosco et al., 2010; Nicaise et al., 2012, 2011). Specifically, open spaces or fields (Cosco et al., 2010; Nicaise et al., 2011) and paths (Nicaise et al., 2012) are salient locations for promoting total physical activity. Research has also demonstrated that physical activity levels vary by the type of activity in which children engage (Brown et al., 2009; Nicaise et al., 2011), with ball/object and wheeled toy play eliciting the most total physical activity in prior studies (Brown et al., 2009; Nicaise et al., 2011). More research into the environment's impacts on schoolyard physical activity during childcare is warranted due to the small number of studies and variability among schoolyards. Differences in methodological approach also make comparisons across studies difficult.

Physical activity measurement

There are many ways to measure participation in physical activity. The gold standard measure of physical activity is direct observation, in which researchers record the activity level of either a group of or individual children, traditionally in real-time. Additionally, environmental variables, like location, activity type, social interaction, or supervision, can be recorded to provide important context to the behavior.

In the physical activity literature, one of the most commonly-used direct observation tools in the preschool population is the Observational System for Recording Physical Activity in Children-Preschool Version (OSRAC-P) (Clevenger et al., 2020b; Truelove et al., 2018), which involves following 1 focal child (discreetly) wherever they travel on the schoolyard (Brown et al., 2006). The OSRAC-P includes, in addition to physical activity level, 12 outdoor context codes referred to as 'activity type' in the current investigation: ball and object, fixed equipment, game, open space, pool or water play, portable equipment, sandbox, snacks, sociodramatic props, teacher arranged gross motor, time out, and wheeled object (Brown et al., 2006). These codes indicate the type of activity (i.e., the behavior in which a child is engaged), not the location, so a child could be in a grassy field or on a concrete basketball court playing with a ball and this would be recorded in both situations as 'ball and object.' Alternatively, the behavior mapping approach relies on scans of target areas, after which the physical activity intensity of individual observed children are recorded on a handheld device (Cosco et al., 2010; Wishart et al., 2019), and the approximate location of the observed child is recorded on a paper map. While direct observation can provide valuable context to physical activity measurement (e.g., activity type or location), there is substantial observer burden (i.e., time, focus) and data are missed when researchers are recording data or not observing a particular child or area (Cox et al., 2020; Howe, Clevenger, Plow, Porter & Sinha, 2018; Oliver, Schofield & Kolt, 2007; Pate, O'Neill & Mitchell, 2010).

Alternatively, monitor-based approaches are capable of capturing continuous data for each child throughout the entire recess period with less researcher burden (Cliff, Reilly & Okely, 2009). For these reasons, unobtrusive, monitor-based approaches have risen in popularity with researchers but have historically not provided context, like activity location. Monitor-based approaches, like the combination of Global Positioning Systems receiver units (GPS) and accelerometers or heart rate monitors, have been used in children and adolescents (Andersen et al., 2015; Clevenger et al., 2020; Clevenger, Sinha & Howe, 2018; Fjørtoft et al., 2009; Fjørtoft, Löfman & Thorén, 2010; Pawlowski, Andersen, Troelsen & Schipperijn, 2016). This approach may provide a valuable surrogate to more established direct observation approaches, like behavior mapping or the OSRAC-P, as they can be used to collect data on a large-scale with many children (with varying ages, sexes, or motor skills) in a variety of childcare programs differing in schoolyard design. However, a limitation of this type of monitor-based approach is that physical location (i.e., latitude and longitude) is often the only type of context measured. As there are multiple afforded activities within each schoolyard location (e.g., both ball/object play and running games are afforded in grassy fields), information may be missed when only measuring location (Kyttä, 2002). However, activity type- and location-based approaches to characterizing outdoor play have not been directly compared.

Rationale for the present study

The commonly-used OSRAC-P protocol is used to measure physical activity by activity type, which may be thought of as an indication of actualized affordances (Wishart et al., 2018). In contrast, behavior mapping and previously-used monitor-based approaches can be used to measure physical activity by schoolyard location, which may be more indicative of potential affordances. As we know that children do not actualize all potential affordances, it theoretically makes sense that activity type- and location-based approaches may produce different findings. As researchers can elect to use either approach, but rarely measure both constructs (Dyment, Bell & Lucas, 2009; Farley, Meriwether, Baker, Rice & Webber, 2008; Maxwell, Mitchell & Evans, 2008; Nicaise et al., 2011, 2012; (Wishart et al., 2019)), understanding the similarities and differences between location- and activity type-based methodologies may inform future research methodology and will help researchers understand if and how comparisons between studies using different approaches can be made.

No studies have directly compared schoolyard location with activity type as measured in the OSRAC-P, a method often used by physical activity researchers (Brown et al., 2009; Howie et al., 2013; Hustyi et al., 2012; Larson, Normand, Morley & Hustyi, 2014; Nicaise et al., 2012, 2011; Pate et al., 2008). However, a study by Nicaise et al. (2011) illustrates the potential difference between activity type and location at 2 schoolyards using a modified version of the OSRAC-P, in which they coded children's location and their activity type. At the first schoolyard, time spent in the sandbox area (221 30-s intervals) and time spent engaging in sand play (212 30-s intervals) was similar, suggesting that children were spending the majority of sandbox time playing in the sand. However, on the second schoolyard, children spent disproportionally more time in the sandbox area (791 30-s intervals), but only 214 30-s intervals were spent engaging in sand play. Thus, children were doing something else while located in the sandbox. The descriptions and sizes of the 2 sandboxes are identical, and their research design did not allow for the identification of what other activities children were engaged in while located in the sandbox. Overall, this indicates that some caution should be exercised when measuring physical activity by location without accompanying activity type (Nicaise et al., 2011), but the extent to which results from these 2 approaches can be compared is unknown.

The overarching purpose of this study was to compare a location- vs activity type-based approach to characterizing children's schoolyard physical activity. Our specific aims were to: (1) identify the percent of preschoolers' time spent in various activity types within schoolyard locations; (2) descriptively compare physical activity level while participating in activity types in different schoolyard locations; and (3) compare physical activity level across either schoolyard locations or activity types to understand how study results would be impacted by the methodological approach.

Our corresponding hypotheses were that: (1) children would participate in multiple activities within each schoolyard location, but a majority of time within a location would be spent on 1 predominant activity (e.g., fixed equipment play while on the fixed equipment); (2) physical activity level for a given activity type would vary by the location in which the activity occurred because locations may afford more or less active versions of the same activity type (e.g., running may occur at a higher speed in open spaces compared to more confined areas, like fixed equipment); and (3) there would be differences in physical activity level by either location or activity type, but the model fit would not be significantly different when using location or activity type as fixed effects.

Materials and methods

The Michigan State University Institutional Review Board approved this study protocol before school/participant recruitment.

2.1. Participants

Three Mid-Michigan preschools (licensed childcare centers) were recruited to participate in this study via phone/email contact. Children 2–5 years of age were eligible to participate. Parents completed a consent form and verbal assent was provided by fifty children (28 girls, 3–5 years of age) from 8 classrooms. Multiple classes participated at each school- 2 classes participated in schools 1 and 2, and both classes used the same schoolyard. Three classes participated in the third school, but 1 class used a different schoolyard. The 4 schoolyards are described in Supplementary Table 1 and illustrated in Supplementary Figures 1–4.

2.2. Setting

Before data collection, each of the 4 schoolyards was visited in person to identify schoolyard locations, defined as an area with a variety of materials that is delineated in some way, often by its physical boundaries or surface type (Sanoff, 1995). On the day of data collection, information about the weather (e.g., temperature), available equipment (e.g., the teacher brought out additional sociodramatic props), and the number of children present was recorded (Supplementary Table 2). Briefly, there were 9–31 children on the schoolyards at a time, and temperature ranged from 19.4 to 29.4 Celsius (66.9–84.9 Fahrenheit). All schoolyards had fixed equipment, open space, and seating, while half of the schoolyards had sandboxes or paths, and 1 schoolyard had a sensory area (water table and mud kitchen). Teachers provided a variety of portable/loose equipment.

2.3. Data collection

Before data collection for this study, this protocol was piloted at a number of schools to ensure feasibility of data collection.

Video observation

For 1 outdoor period per class, the schoolyard was video-recorded with 3 GoPro Hero Session cameras (San Mateo, CA, USA), collecting video at 1080p using a wide-angle view at 30 frames/s. Video camera times were synced to the same cell phone using the GoPro mobile application, and video cameras were started simultaneously using a GoPro Smart Remote. Start and stop times of each video were verified using the GoPro studio desktop application. Cameras were placed to cover the entire schoolyard area, with some overlap to ensure that children could be viewed continuously. While many schools have video surveillance systems in place, privacy is an essential consideration for this study protocol. Parents at participating schools had previously agreed to photo/video capture of their child(ren), and all videos were stored on a protected drive in a locked office and only reviewed by approved research staff to ensure the privacy of children and teachers.

Location and activity type of each child was coded continuously via video observation using Behavioral Observation Research Interactive Software (BORIS; version 7, Torino, Italy) (Friard & Gamba, 2016), which allowed for all 3 videos to be time-aligned and coded for both location and activity type simultaneously (Cox et al., 2020; Howe et al., 2018). Activity type was coded as the 'outdoor activity contexts' as defined in the OSRAC-P, a reliable (Brown et al., 2006) direct observation system that has been used as a criterion measure of preschool's behavior (Dobell et al., 2019; Kahan, Nicaise & Reuben, 2013; Van Cauwenberghe, Gubbels, De Bourdeaudhuij & Cardon, 2011). For example, 'teacher arranged' activity was coded 'when the teacher has planned a gross-motor activity, has arranged the space (with or without materials), is leading the activity, and remains an active participant in the activity' (Brown, 2012). Coding rules as outlined in the OSRAC-P manual (Brown, 2012), were followed, including the hierarchal rules (e.g., if a child is participating in both ball/object and sociodramatic prop play, 'ball/object' is coded as it is of a higher order). However, in 1 schoolyard, the sandbox contained a large kitchen area; while the OSRAC-P mandates that all play in a sandbox be coded as sand play, we coded this behavior as 'sociodramatic' as children were standing on, but not interacting with, sand. While some locations and activity types have identical names (e.g., fixed equipment, open spaces), it is important to note the distinction. For example, being located on fixed equipment simply means one is physically located on that equipment while participating in the fixed equipment activity type indicates that a child is actively using features of the fixed equipment and/or is not participating in some other type of activity (e.g., reading a book).

Time-aligned videos were watched simultaneously, and every time a focal child changed location or activity type, the location was coded by pressing the assigned key (e.g., 'g' for grass) and the activity type was chosen from a drop-down list of options as a modifier to this location code. The software records the start (onset) time for the new behavior and the stop (offset) time for the previous behavior, and this is used to calculate the duration of each event. Each child was coded individually, allowing for continuous coding of location and activity type throughout the entire outdoor period (i.e., no missing data where children were not observed and coded). Observers trained using the OSRAC-P manual and practice videos under the guidance of 1 of the original creators of the direct observation system. One observer, who has been conducting similar video direct observation for several years, coded all videos, while a second trained observer coded a subset of data (~25% of the sample), and second-by-second agreement among observers was 81.5%.

Measurement of physical activity

For the outdoor period, children wore an Actigraph wGT3X-BT accelerometer (ActiGraph LLC, Pensacola, FL) at the right hip, secured by an elastic belt. ActiGraph accelerometers, the most commonly used in studies of preschool children (O'Brien et al., 2018), measure raw acceleration (g) in 3 planes (vertical, anteroposterior, and mediolateral), which is then rectified, filtered (to attenuate non-human movement), and summed (as counts) over a specified period (epoch) (Chen & Basset, 2005; John & Freedson, 2012). Monitors were initialized to measure raw acceleration at a sampling rate of 30 Hz, and data were downloaded as re-integrated counts using ActiLife software (version 6.13.3). Physical activity level was indicated by vector magnitude (the square root of the sum of the squared activity counts from each axis), which was calculated for each 1-s epoch. The vector magnitude metric was selected to decrease the influence of monitor orientation (as data from all axes are combined) (Aadland & Ylvisåker, 2015; Ozemek, Kirschner, Wilkerson, Byun & Kaminsky, 2014), which can be helpful when participants (e.g., young children) have difficulty keeping the monitor belt positioned directly over the right hip.

Data analysis

Direct observation and accelerometer data were matched on a second-by-second basis based on the time stamp in R (R Core Development Team, Vienna, Austria). For descriptive analysis, data were aggregated to the group level as the purpose of this study was not to assess individual behavior, and this is similar to prior direct observation research (Brown et al., 2009). As each schoolyard had a unique design, we collapsed locations into several common locations (e.g., pieces of fixed equipment were all recoded as fixed equipment). Cramér's V was used to assess the strength of the association between location and activity type (Cramér,1946).

To describe the percent of preschoolers' time spent in various activity types within schoolyard locations (aim 1), percent of time in each location spent in each activity type was calculated. To describe physical activity level for each activity type across locations (aim 2), physical activity level for each activity type within each location was calculated. To compare physical activity level across locations or activity types (aim 3), 2 linear mixed models were fit using the lme4 package in R, with location or activity type as the fixed effect and participants nested within schools as random effects with varying intercepts, with physical activity level (vector magnitude counts/s) as the dependent variable. Additionally, a null model with only random effects was fit for comparison. Pairwise differences in physical activity level between locations or activity types were examined using Tukey post hoc tests in the emmeans package, with significance set as $P < 0.01$ (Lenth, 2018). The model fits were compared overall using Vuong's likelihood ratio tests (Vuong, 1989). Of note, future studies should include covariates (e.g., motor skills, age) to assess the impact of personal characteristics on physical activity level across activity types or locations but this was not the purpose of the present study. The purpose of the described analysis is to identify whether model fit is different when using location versus activity type as the fixed effect.

Results

Children provided, on average, 42.9 ± 32.6 min of outdoor data. In school 3, the recorded outdoor period was their only outdoor time for the day. However, in the other 2 schools, this was one of multiple daily outdoor periods (randomly chosen by the primary investigator).

Cramér's V was 0.625, indicating an association between location and activity type. Percent of time spent in each location and percent of time spent in each location in each activity type are found in Table 1. The majority of time was spent in the open spaces location (48.9%) and activity type (37.6%). Less than 5% of the time was spent in seating or sensory locations and the wheeled toys, teacher arranged, portable equipment, and other activity types. Multiple activities occurred within each location, with a median of 5 types of activities occurring within each location. While 94.5% of the time in the sandbox was spent participating in sand play, other locations were more diverse; for example, the majority of time on the path was spent using sociodramatic props (52.7% of the time in this location), but children also participated in open space, wheeled toy, and ball/object activities for 27.9, 17.6, and 1.7% of time, respectively.

Model coefficients can be found in Table 2. While both mixed models (with either location or activity type as the fixed effect) performed better than the null model, there was no difference in model fit between the location or activity type model. Pairwise

Table 1
Percent of time within each schoolyard location spent in each activity type.

Schoolyard location	Activity type: Open space	Socioprops	Fixed equipment	Ball or object	Sandbox	Wheeled toys	Teacher arranged	Portable equipment	Other	Overall
Open Space	63.4	15.1	–	14.4	–	1.3	2.8	2.3	0.8	48.9
Fixed Equipment	–	18.5	66.9	7.7	–	5.7	1.2	–	–	29.0
Path	27.9	52.7	–	1.7	–	17.6	–	–	–	10.8
Sandbox	–	5.5	–	–	94.5	–	–	–	–	5.8
Seating	81.7	16.6	–	0.2	–	0.8	–	–	0.6	4.3
Sensory	10.0	78.9	–	–	0.1	1.0	–	2.3	7.7	1.3
Overall	37.6	20.5	19.4	9.5	5.5	4.2	1.7	1.1	0.5	

Locations and activity types may use the same name (e.g., sandbox) but locations are coded based on the child's physical location on the schoolyard while activity type is coded based on the type of activity in which the child is engaged (e.g., playing with portable equipment).

Table 2
Results of 3 mixed models for physical activity level (vector magnitude counts/s), with participant nested within schools as random effects and no (Model 1), location (Model 2), or activity type (Model 3) fixed effects.

	Model 1 random only					Model 2 location					Model 3 activity type			
	Estimate	*SE*	*t*	*p*		*Estimate*	*SE*	*t*	*p*		*Estimate*	*SE*	*t*	*p*
Intercept	42.2	2.3	18.4	<0.001	Intercept	40.8	2.3	17.5	<0.001	Intercept	40.7	2.3	17.6	<0.001
					Loc. 2	6.3	0.5	13.9	<0.001	Type 2	−3.7	0.5	−6.8	<0.001
					Loc. 3	0.5	0.7	0.7	0.482	Type 3	6.6	0.5	13.4	<0.001
					Loc. 4	3.4	1.0	3.5	<0.001	Type 4	5.9	0.7	8.1	<0.001
					Loc. 5	−6.4	0.8	−8.1	<0.001	Type 5	2.6	1.0	2.6	0.008
					Loc. 6	−1.7	2.3	−0.8	0.451	Type 6	2.3	1.1	2.1	0.035
										Type 7	15.5	1.3	11.9	<0.001
										Type 8	−0.4	1.5	−0.3	0.797
										Type 9	3.0	2.7	1.1	0.271

NS = not significant; Locations are: open space (reference), fixed equipment (2), pathway (3), sandbox (4), seating area (5), or sensory area (6); Activity types are: open space (reference), socioprops (2), fixed equipment (3), ball/object (4), sand play (5), wheeled toy (6), teacher arranged (7), portable equipment (8), or other (9). The most commonly observed location and activity type served as the referent group.

Table 3
Pairwise differences in marginal means (± SE) for physical activity level (vector magnitude counts/s) across schoolyard locations.

Location	Physical activity level(Vector magnitude counts/s)Mean ± SE	95% Confidence interval	Pairwise differences
1. Fixed equipment	47.1 ± 2.4	42.5–51.7	3,4,5,6
2. Sandbox	44.2 ± 2.5	39.3–49.1	4,6
3. Pathway	41.3 ± 2.4	36.6–46.0	1,6
4. Open space	40.8 ± 2.3	36.2–45.3	1,2,6
5. Sensory area	39.1 ± 3.3	32.7–45.4	1
6. Seating area	34.4 ± 2.4	29.7–39.2	1,2,3,4

Means are marginal means; Pairwise differences are significant at $P < 0.01$.

differences illustrate the significant differences in physical activity level between locations (Table 3) and between activity types (Table 4). Of note, the fixed equipment location (47.1 counts/s) resulted in a higher physical activity level compared to paths (−5.8 counts/s), open spaces (−6.3 counts/s), sensory areas (−8.0 counts/s), or seating areas (−12.7 counts/s), while the seating location (34.4 counts/s) resulted in significantly lower physical activity level compared to fixed equipment (+12.7 count/s), sand (+9.8 counts/s), path (+6.9 counts/s), open spaces (+6.4 counts/s). The teacher arranged activity type (56.2 counts/s) resulted in a greater physical activity level compared to all other activity types (−8.9 to 19.1 counts/s). Play with sociodramatic props elicited a lower physical activity level compared to all other activity types (+3.3 to 19.1 counts/s), except the other category.

Table 5 outlines the physical activity level in activity types within different locations. For example, open space activities elicited a vector magnitude of 40.9, 60.7, 35.0, or 88.4 counts/s depending on the location- open spaces, path, seating, or sensory, respectively. Similarly, ball/object activities ranged in vector magnitude from 10.5 counts/s (seating area) to 61.4 counts/s (path) and wheeled toy activities ranged in vector magnitude from 0.0 counts/s (sensory location) to 54.9 counts/s (fixed equipment).

Discussion

The present study aimed to compare physical activity by activity type (as operationalized by the OSRAC-P) and by schoolyard location with the overarching goal of informing the methodology and interpretation of future research in this area. We report that the 2 approaches are not interchangeable, but rather, provide cumulative information. However, some information was consistent between methodological approaches (e.g., most common location/activity type), and our findings suggest that using either location or activity type results in a similar model fit for statistically predicting physical activity level. Thus, the best approach or the need to use a combination of approaches will depend on the research question, a theme we will discuss further below.

Table 4
Pairwise differences in marginal means (± SE) for physical activity level (vector magnitude counts/s) across activity types.

Activity type	Physical activity level (Vector magnitude counts/s) Mean ± SE	95% confidence interval	Pairwise differences
1. Teacher arranged	56.2 ± 2.6	51.1–61.4	2,3,4,5,6,7,8,9
2. Fixed equipment	47.3 ± 2.3	42.7–51.9	1,5,6,7,8,9
3. Ball/object	46.6 ± 2.4	41.9–51.3	1,7,8,9
4. Other	43.7 ± 3.6	36.8–50.7	1
5. Sand play	43.4 ± 2.5	38.5–48.2	1,2,9
6. Wheeled toy	43.1 ± 2.5	38.1–48.0	1,2,9
7. Open space	40.7 ± 2.3	36.2–45.3	1,2,3,9
8. Portable equipment	40.4 ± 2.7	35.0–45.7	1,2,3
9. Socioprops	37.1 ± 2.4	32.5–41.7	1,2,3,5,6,7

Means are marginal means; Pairwise differences are significant at $P < 0.01$.

Table 5
Physical activity level (as mean vector magnitude counts/s) in each activity type within each schoolyard location.

Activity types within locations	Physical activity level (Vector magnitude counts/s)
Open space	
Open space	40.9
Socioprops	41.4
Ball/object	49.7
Wheeled toys	36.5
Teacher arranged	46.8
Portable equipment	26.5
Other	39.8
Fixed equipment	
Socioprops	38.1
Fixed equipment	44.7
Ball/object	34.0
Wheeled toys	54.9
Teacher arranged	74.4
Path	
Open space	60.7
Socioprops	32.2
Ball/object	61.4
Wheeled toys	30.6
Sandbox	
Socioprops	42.6
Sandbox	37.6
Seating	
Open space	35.0
Socioprops	21.5
Ball/object	10.5
Wheeled toys	30.3
Other	26.5
Sensory	
Open space	88.4
Socioprops	30.1
Sandbox	3.5
Wheeled toys	0.0
Portable equipment	91.9
Other	83.7

Locations and activity types may use the same name (e.g., sandbox) but locations are coded based on the child's physical location on the schoolyard while activity type is coded based on the type of activity in which the child is engaged (e.g., playing with portable equipment).

Activity types within locations

Our first aim was to identify time spent in each activity type within different schoolyard locations. We supported our hypothesis that children participated in multiple activities within each location, which is in line with prior research by Nicaise et al. (2012), and the theory of affordances (Bjørgen, 2016; Kyttä, 2004; Storli & Hagan, 2010; Wyver & Little, 2018). While Nicaise et al. (2012) did not report the percent of time spent in various activity types within locations, they reported the most commonly observed activity types in each location. In contrast to our findings (Table 1), they reported primarily wheeled toy use on the cement path and ball/object, open space, and teacher-arranged activities on the playground and in the grass. These differences may be due to methodological differences (e.g., how locations were defined), differences in available equipment (e.g., provision of sociodramatic props) or because Nicaise et al. (2012) included data from both pre- and post-intervention, which they reported altered the types of activities in which children engaged within locations on the schoolyard. Both the present study (4 schoolyards) and the study by Nicaise et al. (2012) (1 schoolyard pre- and post-intervention) included only a limited number of schoolyards, but both support that locations afford multiple activity types.

Despite participating in multiple types of activities within each location, children spent the majority of their time in 1 expected activity type. Both approaches identified open spaces, followed by fixed equipment as the most common locations and activity types. These findings are in line with previous location-based studies (Cosco et al., 2010; Nicaise et al., 2012, 2011; Wishart et al., 2018) and studies using an activity type-based approach (Brown & Burger, 1984; Brown et al., 2009; Howie et al., 2013; Soini et al., 2014). This congruency between approaches illustrates that the overall findings as to what children 'do' during outdoor time are similar, regardless of the methodological approach used.

Some activity types (e.g., fixed equipment, sand play) only occurred in a finite number of locations, which was to be expected because these types of play are afforded in these locations. For these activity types, results from location-based and activity type-based approaches are comparable. Conversely, other activity types occurred in multiple locations. For example, play with sociodramatic props took place in all 6 locations (i.e., sociodramatic play is afforded in all locations), so only recording that children participated in play with sociodramatic props (as in the OSRAC-P) disregards information about where this behavior occurred. Similarly, information is missed when only location is considered, as multiple activity types can occur within a location. For example, the fixed equipment location encompassed fixed equipment, sociodramatic, ball/object, teacher arranged, and wheeled toy activity types, but the most prevalent activity type was, as expected, fixed equipment play (66.9%). Research using location-only approaches (e.g., Andersen et al., 2015; Clevenger et al., 2018; Fjørtoft et al., 2009, 2010; Pawlowski et al., 2016) would not be able to readily capture these differences in activity type within schoolyard locations without collecting or using some additional information.

Our finding that activity type and location are not equivalent is supported by previous research by Nicaise et al., 2011 who used a modified version of the OSRAC-P to code both location and activity type (called context). While they did not report the activity types in which children participated within each location or physical activity level by location or by activity type across locations, their results suggested that there was a disparity between time in locations and time spent in the corresponding activity type. For example, in 1 school, 604 intervals were spent on the asphalt cycle path, but only 166 intervals were spent using wheeled toys. Thus, there were at least 438 observation intervals in which a child was located on the asphalt cycle path, but was not using a wheeled toy (Nicaise et al., 2011). This difference is in line with our findings that suggest that much of the time in the path location is spent participating in other types of activities (e.g., open space). However, we also report that much of the time spent in that activity type (wheeled object) does not occur on the cycle path (e.g., children may take the wheeled toys into the grass). Thus, a location can afford multiple activity types, and activity types can be afforded in multiple locations.

Physical activity by activity type across locations

The disconnect between the intended behavior and actualized affordances may indicate the ineffectiveness of design (Refshauge, Stigsdotter, Lamm & Thorleifsdottir, 2015), but may also have implications for interventions. It is currently recommended that childcare programs provide children with a path for wheeled toys (McWilliams et al., 2009; Spencer & Wright, 2014), and the presence of paths for wheeled toys is an item included in child care quality scales (Benjamin et al., 2007; Early, Sideris, Neitzel, LaForett & Nehler, 2018). While the presence of a path/track (Gubbels, Van Kann & Jansen, 2012) and wheeled toys (Brown et al., 2009) have both been associated with physical activity participation, we reported the lowest levels of physical activity when children participated in wheeled toy play on the track, a disconnect supported by Sando (2019). Instead, children seemed to be more active when participating in open space or ball/object activities (e.g., running games) in this location. Children were also relatively active when using wheeled toys on fixed equipment (although we recognize that this may be unsafe). As researchers have not investigated where wheeled toy use occurs or reported the physical activity levels elicited during specific types of activities that take place within the path location, the assumption that wheeled toy activities occur on tracks (Soini et al., 2014) should not be made. Future research may investigate the relative importance of the location (path) versus the piece of equipment (activity type) so that program directors and interventionists can effectively allocate limited funds and resources (Anthamatten et al., 2014). Our findings suggest that the addition of a path that only affords wheeled object use may not promote increases in physical activity, but this needs to be studied in larger samples before conclusions may be drawn. Of note, only a small number of schools had wheeled objects available in the present study, and children at those schools were generally younger and, therefore, may not have been as physically active while engaging with wheeled toys.

The finding that physical activity level varied by location in which a type of activity occurred is in line with affordance theory. For example, while children may be holding a ball/object in any location, a seating area does not afford active ball/object play (e.g., throwing), and this is reflected in an average physical activity level of 10.5 counts/s while participating in this type of activity in this location. Conversely, a path (61.4 counts/s) or open space (49.7 counts/s) may more readily afford throwing or batting. This variability could explain why intervention studies providing additional portable equipment have been both successful (Hannon & Brown, 2008) and unsuccessful (Cardon, Labarque, Smits & De Bourdeaudhuij, 2009) in promoting physical activity in preschool-aged children, as there may not be adequate space in which to use these items for the intended type of play. Researchers could assess activity type by location following this type of intervention. Additionally, more research may be needed on specific types of ball/object equipment (i.e., there may be differences between balls versus hula-hoops) or more specific activity types (e.g., throw versus standing with an object in hand). Overall, the finding that physical activity varies by activity and location supports that measuring both activity type and location may be useful in future studies.

Physical activity by location or activity type

Another aim of the present study was to compare physical activity level (vector magnitude counts/s) between schoolyard locations and activity types to understand how the chosen methodological approach would impact study outcomes. Not all locations and activity types are directly comparable, but there are 2 findings of note. First, physical activity level is similar between locations and comparable activity types. For example, fixed equipment (47.1 vs 47.3 counts/s), sandbox (44.2 vs 443.4), and open space (40.8 vs 40.7) resulted in similar physical activity levels. Second, the conclusions that would be drawn from the 2 methodological approaches are slightly different, despite similar model fit. Results from the location-based approach indicate that fixed equipment is a location that affords physical activity, while seating areas are associated with lower physical activity levels, both findings that are supported by previous research (Cosco et al., 2010; Hustyi et al., 2012; Larson et al., 2014; Smith et al., 2014). Results from the activity type-based approach indicate that participating in teacher arranged activities or fixed equipment play stimulates physical activity, which are findings also supported by prior research (Brown et al., 2009; Howie et al., 2013). As teacher arranged activity only occurred for 2.5% of observation time (similar to the 2.6% reported by Brown et al., 2009), it is likely a trivial issue that location-based approaches do not inherently collect this information. However, future methodological advancements could be developed (e.g., use of Bluetooth proximity tagging) to capture this type of activity. Researchers should consider their research question when choosing a methodological approach.

Methodological considerations for future research

Use of only 1 type of approach for characterizing physical activity (activity type- or location-based) may be appropriate based on the research question and study resources, while a combination of both approaches may be needed if the study requires a more nuanced understanding of children's behavior. Future studies using activity type-based approaches, like the OSRAC-P, may use a modified version that includes location like that used by Nicaise et al. (2011) and Nicaise et al. (2012). Future research using location-based approaches may be able to better capture some of the variability in activity type within locations by including teacher reports of additional provided equipment (e.g., chalk). For example, in the present study, it may be surprising that the most common activity type occurring on paths was sociodramatic props, but this was because children used chalk to draw on the path for much of their outdoor time at 1 schoolyard (schoolyard 4). In contrast, at another school (schoolyard 2), children had access to wheeled toys, like tricycles, that only a small number of children chose to access. In another school, children were provided with animal masks during 1 observation, so they were coded as participating in sociodramatic props for much of their fixed equipment time. Alternatively, at the same schoolyard, but during a different observation, no additional equipment was provided to children. Collecting information on the equipment provided to children may provide additional context when only measuring children's location.

Researchers using location-based approaches (e.g., behavior mapping) may also elect to record information about activity type (Wishart et al., 2018). Additionally, researchers that have used monitor-based location-based approaches (e.g., Global Positioning Systems plus accelerometry) have not used all available monitor information (H. B. Andersen et al., 2015; Clevenger et al., 2018; Fjørtoft et al., 2009, 2010; Pawlowski et al., 2016), but some types of activities may be able to be recognized using machine learning algorithms informed by speed, body angle, or acceleration-based metrics (Hagenbuchner et al., 2015). For example, wheeled riding toys could be discerned from open space activities while a child is located on the track based on speed and posture (sitting vs standing). While the development of such algorithms is outside the scope of this paper, we demonstrate that classifying activity type, in addition to schoolyard location, may be important in preschool-aged samples during outdoor time.

Overall, researchers should consider the pros and cons of location-based versus activity type-based approaches. Observational scales like the OSRAC-P (Brown et al., 2006), the System for Observing Play and Leisure Activity in Youth (SOPLAY; McKenzie, 2006), and the System for Observing Children's Activity and Relationships during Play (Ridgers, Fairclough & Stratton, 2010) are reliable and when using video observation, may be able to provide continuous, individual-level data on the context of play behaviors. The OSRAC, and similar scales, also allow for comparable measurements between schoolyards, despite differences between schools in the types of specific activities in which children engage (e.g., multiple types of play are collapsed to 1 sociodramatic play code). Conversely, using a location-based approach may also be amenable to assessing a variety of playgrounds as the locations of interest can vary from playground to playground. While we collapsed the data into common locations for clarity in the present paper, we were able to identify time spent using specific pieces of, for example, fixed equipment (e.g., spinning net, swings). This approach aligns with recommendations by Pellegrini, (1987) that children's behavior should be explored at a micro-level, then aggregated into macro-level categories to allow for ethological variability between children and schools. Being able to measure locational differences may also be important for recognizing what children use on the schoolyard (Clevenger et al., 2019; Lucas & Dyment, 2010; Pawlowski, Andersen, Arvidsen & Schipperijn, 2019) and to inform (H. B. Andersen et al., 2015; Derr and Rigolon, 2016) or assess (H. B. Andersen et al., 2015; Nicaise et al., 2012; Raney et al., 2019) schoolyard redesign. For example, Nicaise et al. (2012) removed some small pieces of fixed equipment, which increased the amount of open space and subsequently increased physical activity in 1 preschool. Combining measures of both location and activity type could identify if children used these structures before their removal and explain why this intervention was effective for promoting physical activity participation.

Limitations of the present study

It is important to note that there are limitations to this study. These findings are not meant to provide a representative indication of where children spent their outdoor time, as we used a small number of children and schools and only 1 outdoor period. We noticed that types of teacher involvement and portable or sociodramatic equipment brought to the playground varied substantially between outdoor periods, both within and between days (not shown), and this impacted the types of activities children participated in within the schoolyard locations. Thus, our findings may not represent where children habitually spend their outdoor time. However, our goal was to encompass a variety of schoolyard locations and types of equipment to indicate the number and types of activities in which children participate while in various locations. Future research should include covariates to explore differences between schoolyards and sub-groups (e.g., by motor skill, weight status). For example, there is evidence in older children that there are differences between sexes in physical activity within schoolyard locations and this may be due to children of different sexes participating in different types of activities within locations (Black, Menzel & Bungum, 2015; Dyment et al., 2009). A strength of the present study was the use of continuous video direct observation, which allowed us to review and code multiple behaviors second-by-second for each child.

Implications for researchers and practitioners

This work has implications for both researchers and practitioners. For researchers, consider the aim of a study when selecting the methodological approach. If research aims to characterize schoolyard physical activity (e.g., overall, by sex) or pre/post-intervention, it would be ideal to measure both activity type and location, and future research should aim to develop methodologies for this purpose. However, if this is not feasible, consider the advantages and disadvantages of each approach and understand the limitations of the chosen approach. For example, if measuring activity intensity by location, there will be variability within locations due to participation in different types of activities. If measuring activity intensity by activity type, there will be variability across locations.

For practitioners, this work suggests that children participate in a variety of activities in locations across the schoolyard, and sometimes participating in an 'unexpected' type of activity drives physical activity. So, when it can safely be done, allow children to participate in a variety of activities within locations. For example, we found children were highly active when participating in open space or ball/object activities on the path. This behavior may not be how this equipment or location was meant to be used, but this highlights the fact that the actualized affordances in a location on the schoolyard may not be what a researcher, teacher, or schoolyard designer initially intended.

Conclusions

We can conclude that there is some discrepancy between activity type and location, but the overall patterns identified in children's behavior were similar for both approaches. For example, open spaces were the most prevalent activity type and the location in which children spent the majority of their time, and both approaches supported that fixed equipment facilitates physical activity. These consistencies suggest that studies focusing on activity type (e.g., OSRAC-P) may be comparable to those focusing on location, but caution should be exercised for some activity types or locations. However, schoolyard location and activity type seem to provide cumulative information that together may better inform our understanding of children's behavior during provided outdoor time. Thus, it is not how *or* where children are playing that impacts physical activity participation, but how *and* where they are playing.

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Disclosure

No potential conflict of interest was reported by the authors.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ecresq.2021.03.012.

References

Aadland, E., & Ylvisåker, E. (2015). Reliability of the Actigraph GT3X+ accelerometer in adults under free-living conditions. *PloS one, 10*(8), Article e0134606.

Andersen, H. B., Klinker, C. D., Toftager, M., Pawlowski, C. S., & Schipperijn, J. (2015a). Objectively measured differences in physical activity in five types of schoolyard area. *Landscape and Urban Planning, 134*, 83–92.
Andersen, H. B., Pawlowski, C. S., Scheller, H. B., Troelsen, J., Toftager, M., & Schipperijn, J. (2015b). Activating schoolyards: Study design of a quasi-experimental schoolyard intervention study. *BMC public health, 15*(1), 523.
Anthamatten, P., Brink, L., Kingston, B., Kutchman, E., Lampe, S., & Nigg, C. (2014). An assessment of schoolyard features and behavior patterns in children's utilization and physical activity. *Journal of Physical Activity and Health, 11*(3), 564–573.
Benjamin, S. E., Ammerman, A., Sommers, J., Dodds, J., Neelon, B., & Ward, D. S. (2007). Nutrition and physical activity self-assessment for child care (NAP SACC): Results from a pilot intervention. *Journal of Nutrition Education and Behavior, 39*(3), 142–149.
Berg, S. (2015). Children's activity levels in different playground environments: An observational study in four Canadian preschools. *Early Childhood Education Journal, 43*(4), 281–287.
Bjørgen, K. (2016). Physical activity in light of affordances in outdoor environments: Qualitative observation studies of 3–5 years olds in kindergarten. *SpringerPlus, 5*(1), 950.
Black, I. E., Menzel, N. N., & Bungum, T. J. (2015). The relationship among playground areas and physical activity levels in children. *Journal of Pediatric Health Care, 29*(2), 156–168.
Bower, J. K., Hales, D. P., Tate, D. F., Rubin, D. A., Benjamin, S. E., & Ward, D. S. (2008). The childcare environment and children's physical activity. *American Journal of Preventive Medicine, 34*(1), 23–29.
Broekhuizen, K., Scholten, A. M., & De Vries, S. I. (2014). The value of (pre) school playgrounds for children's physical activity level: A systematic review. *International Journal of Behavioral Nutrition and Physical Activity, 11*(1), 59.
Brown, J. G., & Burger, C. (1984). Playground designs and preschool children's behaviors. *Environment and Behavior, 16*(5), 599–626.
Brown, W. (2012). Observational System for Recording Physical Activity in Children- Preschoolers (OSRAC-P) Training Manual for Observers. http://www.asph.sc.edu/USC_CPARG/pdf/OSRAC_Manual.pdf Accessed April 2021.
Brown, W. H., Pfeiffer, K. A., McIver, K. L., Dowda, M., Addy, C. L., & Pate, R. R. (2009a). Social and environmental factors associated with preschoolers' nonsedentary physical activity. *Child Development, 80*(1), 45–58.
Brown, W. H., Pfeiffer, K. A., McIver, K. L., Dowda, M., Addy, C. L., & Pate, R. R. (2009b). Social and environmental factors associated with preschoolers' nonsedentary physical activity. *Child Development, 80*(1), 45–58.
Brown, W. H., Pfeiffer, K. A., McIver, K. L., Dowda, M., Almeida, J. M., & Pate, R. R. (2006). Assessing preschool children's physical activity: The Observational System for Recording Physical Activity in children-preschool version. *Research Quarterly for Exercise and Sport, 77*(2), 167–176.
Cardon, G., Labarque, V., Smits, D., & De Bourdeaudhuij, I. (2009). Promoting physical activity at the pre-school playground: The effects of providing markings and play equipment. *Preventive Medicine, 48*(4), 335–340.
Cardon, G., Van Cauwenberghe, E., Labarque, V., Haerens, L., & De Bourdeaudhuij, I. (2008). The contribution of preschool playground factors in explaining children's physical activity during recess. *International Journal of Behavioral Nutrition and Physical Activity, 5*(1), 11.
Chen, K. Y., & David R Bassett, J. R. (2005). The technology of accelerometry-based activity monitors: Current and future. *Medicine & Science in Sports and Exercise, 37*(11), S490–S500.
Clevenger, K. A., Grady, S. C., Erickson, K., & Pfeiffer, K. A. (2020a). Use of a spatiotemporal approach for understanding preschoolers' playground activity. *Spatial and Spatio-temporal Epidemiology, 35*, Article 100376.
Clevenger, K. A., Sinha, G., & Howe, C. A. (2018). Comparison of methods for analyzing Global Positioning System and aaccelerometer data during school recess. *Measurement in Physical Education and Exercise Science, 23*(1), 58–68.
Clevenger, K. A., Wierenga, M. J., Howe, C. A., & Pfeiffer, K. A. (2020b). A Systematic Review of Child and Adolescent Physical Activity by Schoolyard Location. *Kinesiology Review, 9*(2), 147–158.
Cliff, D. P., Reilly, J. J., & Okely, A. D. (2009). Methodological considerations in using accelerometers to assess habitual physical activity in children aged 0–5 years. *Journal of Science and Medicine in Sport, 12*(5), 557–567.
Copeland, K. A., Khoury, J. C., & Kalkwarf, H. J. (2016). Child care center characteristics associated with preschoolers' physical activity. *American Journal of Preventive Medicine, 50*(4), 470–479.
Cosco, N. G. (2006). *Motivation to move: Physical activity affordances in preschool play areas [dissertation]* (pp. 65–66). Edinburgh (Scotland): School of Landscape Architecture, Heriot Watt University.
Cosco, N. G., Moore, R. C., & Islam, M. Z. (2010). Behavior mapping: A method for linking preschool physical activity and outdoor design. *Medicine and Science in Sports and Exercise, 42*(3), 513–519.
Cox, M. F., Petrucci, G. J., Marcotte, R. T., Masteller, B. R., Staudenmayer, J., Freedson, P. S., et al. (2020). A novel video-based direct observation system for assessing physical activity and sedentary behavior in children and young adults. *Journal for the Measurement of Physical Behaviour, 1*, 1–8.
Cramér, H. (1946). A contribution to the theory of statistical estimation. *Scandinavian Actuarial Journal, 1946*(1), 85–94.
Derr, V., & Rigolon, A. (2016). Participatory schoolyard design for health and well-being: Policies that support play in urban green spaces. In C. Freeman, & P. Tranter, & Skelton, Tracey (Eds.), *Risk, Protection, Provision, and Policy* (pp. 145–167). Springer: Singapore.
Dobell, A. P., Eyre, E. L., Tallis, J., Chinapaw, M. J., Altenburg, T. M., & Duncan, M. J. (2019). Examining accelerometer validity for estimating physical activity in pre-schoolers during free-living activity. *Scandinavian Journal of Medicine & Science in Sports, 29*(10), 1618–1628.
Dowda, M., Brown, W. H., McIver, K. L., Pfeiffer, K. A., O'Neill, J. R., Addy, C. L., et al. (2009). Policies and characteristics of the preschool environment and physical activity of young children. *Pediatrics, 123*(2), e261–e266.
Dyment, J. E., Bell, A. C., & Lucas, A. J. (2009). The relationship between school ground design and intensity of physical activity. *Children's Geographies, 7*(3), 261–276.
Early, D. M., Sideris, J., Neitzel, J., LaForett, D. R., & Nehler, C. G. (2018). Factor structure and validity of the Early Childhood Environment Rating Scale–Third Edition (ECERS-3). *Early Childhood Research Quarterly, 44*, 242–256.
Farley, T. A., Meriwether, R. A., Baker, E. T., Rice, J. C., & Webber, L. S. (2008). Where do the children play? The influence of playground equipment on physical activity of children in free play. *Journal of Physical Activity and Health, 5*(2), 319.
Fjørtoft, I., Kristoffersen, B., & Sageie, J. (2009). Children in schoolyards: Tracking movement patterns and physical activity in schoolyards using global positioning system and heart rate monitoring. *Landscape and Urban Planning, 93*(3–4), 210–217.
Fjørtoft, I., Löfman, O., & Thorén, K. H. (2010). Schoolyard physical activity in 14-year-old adolescents assessed by mobile GPS and heart rate monitoring analysed by GIS. *Scandinavian Journal of Public Health, 38*(5 suppl), 28–37.
Friard, O., & Gamba, M. (2016). BORIS: A free, versatile open-source event-logging software for video/audio coding and live observations. *Methods in Ecology and Evolution, 7*(11), 1325–1330.
Gubbels, J. S., Kremers, S. P., van Kann, D. H., Stafleu, A., Candel, M. J., Dagnelie, P. C., Thijs, C., & de Vries, N. K. (2011). Interaction between physical environment, social environment, and child characteristics in determining physical activity at child care. *Health Psychol., 30*(1), 84–90.
Gebel, K., Bauman, A. E., & Petticrew, M. (2007). The physical environment and physical activity: A critical appraisal of review articles. *American Journal of Preventive Medicine, 32*(5), 361–369.
Gibson, J. J. (1977). *The theory of affordances*. Hillsdale, NJ: Lawrence Erlbaum Associates.
Gubbels, J. S., Van Kann, D. H., & Jansen, M. W. (2012). Play equipment, physical activity opportunities, and children's activity levels at childcare. *Journal of Environmental and Public Health, 2012*, 1–8.
Gunter, K. B., Rice, K. R., Ward, D. S., & Trost, S. G. (2012). Factors associated with physical activity in children attending family child care homes. *Preventive Medicine, 54*(2), 131–133.
Hagenbuchner, M., Cliff, D.P., Trost, S.G., Van Tuc, N., Peoples, G.E. (2015). Prediction of activity type in preschool children using machine learning techniques. Journal of Science and Medicine in Sport, 18(4), 426–431.
Hannon, J. C., & Brown, B. B. (2008). Increasing preschoolers' physical activity intensities: An activity-friendly preschool playground intervention. *Preventive Medicine, 46*(6), 532–536.
Heft, H. (1988). Affordances of children's environments: A functional approach to environmental description. *Children's Environments Quarterly, 5*(3), 29–37.
Henderson, K. E., Grode, G. M., O'Connell, M. L., & Schwartz, M. B. (2015). Environmental factors associated with physical activity in childcare centers. *International Journal of Behavioral Nutrition and Physical Activity, 12*(1), 43.
Holmes, R. M. (2012). The outdoor recess activities of children at an urban school: Longitudinal and intraperiod patterns. *American Journal of Play, 4*(3), 327–351.
Holmes, R. M., & Procaccino, J. K. (2009). Preschool children's outdoor play area preferences. *Early Child Development and Care, 179*(8), 1103–1112.
Howe, C. A., Clevenger, K. A., Plow, B., Porter, S., & Sinha, G. (2018). Using video direct observation to assess children's physical activity during recess. *Pediatric Exercise Science, 30*(4), 516–523.
Howie, E. K., Brown, W. H., Dowda, M., McIver, K. L., & Pate, R. R. (2013). Physical activity behaviours of highly active preschoolers. *Pediatric Obesity, 8*(2), 142–149.
Hustyi, K. M., Normand, M. P., Larson, T. A., & Morley, A. J. (2012). The effect of outdoor activity context on physical activity in preschool children. *Journal of Applied Behavior Analysis, 45*(2), 401–405.
John, D., & Freedson, P. (2012). ActiGraph and Actical physical activity monitors: A peek under the hood. *Medicine and Science in Sports and Exercise, 44*(Suppl 1), S86.
Kahan, D., Nicaise, V., & Reuben, K. (2013). Convergent validity of four accelerometer cutpoints with direct observation of preschool children's outdoor physical activity. *Research Quarterly for Exercise and Sport, 84*(1), 59–67.
Kyttä, M. (2002). Affordances of children's environments in the context of cities, small towns, suburbs and rural villages in Finland and Belarus. *Journal of Environmental Psychology, 22*(1), 109–123.
Kyttä, M. (2004). The extent of children's independent mobility and the number of actualized affordances as criteria for child-friendly environments. *Journal of Environmental Psychology, 24*(2), 179–198.
Larson, T. A., Normand, M. P., Morley, A. J., & Hustyi, K. M. (2014). The role of the physical environment in promoting physical activity in children across different group compositions. *Behavior Modification, 38*(6), 837–851.
Lenth (2018).
Lerstrup, I., & Konijnendijk van den Bosch, C. (2017). Affordances of outdoor settings for children in preschool: Revisiting heft's functional taxonomy. *Landscape Research, 42*(1), 47–62.
Lucas, A. J., & Dyment, J. E. (2010). Where do children choose to play on the school ground? The influence of green design. *Education, 38*(2), 177–189.

Maxwell, L. E., Mitchell, M. R., & Evans, G. W. (2008). Effects of play equipment and loose parts on preschool children's outdoor play behavior: An observational study and design intervention. *Children Youth and Environments, 18*(2), 36–63.

McGuire, S. (2012). *Institute of medicine (IOM) early childhood obesity prevention policies* (p. 2011). Washington, DC: The National Academies Press.

McKenzie, T.L. (2006). System for observing play and leisure activity in youth (SOPLAY). https://activelivingresearch.org/sites/activelivingresearch.org/files/SOPLAY_Protocols.pdf Accessed April 2021

McWilliams, C., Ball, S. C., Benjamin, S. E., Hales, D., Vaughn, A., & Ward, D. S. (2009). Best-practice guidelines for physical activity at child care. *Pediatrics, 124*(6), 1650–1659.

Nicaise, V., Kahan, D., Reuben, K., & Sallis, J. F. (2012). Evaluation of a redesigned outdoor space on preschool children's physical activity during recess. *Pediatric Exercise Science, 24*(4), 507–518.

Nicaise, V., Kahan, D., & Sallis, J. F. (2011). Correlates of moderate-to-vigorous physical activity among preschoolers during unstructured outdoor play periods. *Preventive Medicine, 53*(4), 309–315.

O'Brien, K.T., Vanderloo, L.M., Bruijns, B.A., Truelove, S., Tucker, P. (2018). Physical activity and sedentary time among preschoolers in centre-based childcare: a systematic review. International Journal of Behavioral Nutrition and Physical Activity, 15(1), 1-16.

Oliver, M., Schofield, G. M., & Kolt, G. S. (2007). Physical activity in preschoolers. *Sports Medicine, 37*(12), 1045–1070.

Ozemek, C., Kirschner, M. M., Wilkerson, B. S., Byun, W., & Kaminsky, L. A. (2014). Intermonitor reliability of the GT3X+ accelerometer at hip, wrist and ankle sites during activities of daily living. *Physiological Measurement, 35*(2), 129.

Pate, R. R., McIver, K., Dowda, M., Brown, W. H., & Addy, C. (2008). Directly observed physical activity levels in preschool children. *Journal of School Health, 78*(8), 438–444.

Pate, R. R., O'Neill, J. R., Brown, W. H., Pfeiffer, K. A., Dowda, M., & Addy, C. L. (2015). Prevalence of compliance with a new physical activity guideline for preschool-age children. *Childhood Obesity, 11*(4), 415–420.

Pate, R. R., O'NEILL, J. R., & Mitchell, J. (2010). Measurement of physical activity in preschool children. *Medicine & Science in Sports and Exercise, 42*(3), 508–512.

Pawlowski, C. S., Andersen, H. B., Arvidsen, J., & Schipperijn, J. (2019). Changing recess geographies: Children's perceptions of a schoolyard renovation project promoting physical activity. *Children's Geographies, 17*(6), 664–675.

Pawlowski, C. S., Andersen, H. B., Troelsen, J., & Schipperijn, J. (2016). Children's physical activity behavior during school recess: A pilot study using GPS, accelerometer, participant observation, and go-along interview. *PloS one, 11*(2), Article e0148786.

Pellegrini, A. D. (1987). Children on playgrounds: A review of "What's out there". *Children's Environments Quarterly, 4*(4), 2–7.

Piercy, K. L., Troiano, R. P., Ballard, R. M., Carlson, S. A., Fulton, J. E., Galuska, D. A., et al. (2018). The physical activity guidelines for Americans. *JAMA, 320*(19), 2020–2028.

Raney, M. A., Hendry, C. F., & Yee, S. A. (2019). Physical activity and social behaviors of urban children in green playgrounds. *American Journal of Preventive Medicine, 56*(4), 522–529.

Raustorp, A., Pagels, P., Boldemann, C., Cosco, N., Söderström, M., & Mårtensson, F. (2012). Accelerometer measured level of physical activity indoors and outdoors during preschool time in Sweden and the United States. *Journal of Physical Activity and Health, 9*(6), 801–808.

Refshauge, A. D., Stigsdotter, U. K., Lamm, B., & Thorleifsdottir, K. (2015). Evidence-based playground design: Lessons learned from theory to practice. *Landscape Research, 40*(2), 226–246.

Ridgers, N. D., Fairclough, S. J., & Stratton, G. (2010). Variables associated with children's physical activity levels during recess: The A-CLASS project. *International Journal of Behavioral Nutrition and Physical Activity, 7*(1), 74.

Sando, O. J. (2019). The outdoor environment and children's health: A multilevel approach. *International Journal of Play, 8*(1), 39–52.

Sanoff, H. (1995). *Creating environments for young children*. Raleigh, NC: North Carolina State University Available at https://eric.ed.gov/?id=ED394640 .

Sevimli-Celik, S. (2018). Active Designs for Movement in Early Childhood Environments. In *Physical activity and health promotion in the early years* (pp. 225–239). Cham: Springer.

Smith, W. R., Moore, R., Cosco, N., Wesoloski, J., Danninger, T., Ward, D. S., et al. (2014). Increasing physical activity in childcare outdoor learning environments: The effect of setting adjacency relative to other built environment and social factors. *Environment and Behavior, 48*(4), 550–578.

Soini, A., Villberg, J., Sääkslahti, A., Gubbels, J., Mehtälä, A., Kettunen, T., et al. (2014). Directly observed physical activity among 3-year-olds in Finnish childcare. *International Journal of Early Childhood, 46*(2), 253–269.

Spencer, K. H., & Wright, P. M. (2014). Quality outdoor play spaces for young children. *Young Children, 69*(5), 28–34.

Storli, R., & Hagen, T. L. (2010). Affordances in outdoor environments and children's physically active play in pre-school. *European Early Childhood Education Research Journal, 18*(4), 445–456.

Sugiyama, T., Okely, A. D., Masters, J. M., & Moore, G. T. (2012). Attributes of child care centers and outdoor play areas associated with preschoolers' physical activity and sedentary behavior. *Environment and Behavior, 44*(3), 334–349.

Tandon, P. S., Saelens, B. E., Zhou, C., Kerr, J., & Christakis, D. A. (2013). Indoor versus outdoor time in preschoolers at child care. *American Journal of Preventive Medicine, 44*(1), 85–88.

Truelove, S., Bruijns, B. A., Vanderloo, L. M., O'Brien, K. T., Johnson, A. M., & Tucker, P. (2018). Physical activity and sedentary time during childcare outdoor play sessions: A systematic review and meta-analysis. *Preventive Medicine, 108*, 74–85.

Van Cauwenberghe, E., Gubbels, J., De Bourdeaudhuij, I., & Cardon, G. (2011). Feasibility and validity of accelerometer measurements to assess physical activity in toddlers. *International Journal of Behavioral Nutrition and Physical Activity, 8*(1), 67.

Vanderloo, L. M., Tucker, P., Johnson, A. M., & Holmes, J. D. (2013). Physical activity among preschoolers during indoor and outdoor childcare play periods. *Applied Physiology, Nutrition, and Metabolism, 38*(11), 1173–1175.

Vanderloo, L. M., Tucker, P., Johnson, A. M., Van Zandvoort, M. M., Burke, S. M., & Irwin, J. D. (2014). The influence of centre-based childcare on preschoolers' physical activity levels: A cross-sectional study. *International Journal of Environmental Research and Public Health, 11*(2), 1794–1802.

Vuong, Q. H. (1989). Likelihood ratio tests for model selection and non-nested hypotheses. *Econometrica: Journal of the Econometric Society, 57*(2), 307–333.

Wishart, L., Cabezas-Benalcázar, C., Morrissey, A. M., & Versace, V. L. (2019). Traditional vs naturalised design: A comparison of affordances and physical activity in two preschool playscapes. *Landscape Research, 44*(8), 1031–1049.

Wyver, S., & Little, H. (2018). Early childhood education environments: Affordances for risk-taking and physical activity in play. In *Physical activity and health promotion in the early years* (pp. 41–55). Cham: Springer.